The Big Picture

2

Part of becoming senior in an organization is the privilege of running meetings. When I reached this developmental milestone, I soon learned always to prepare a detailed written agenda before each meeting. That preparation forced me think about both the process and possible outcomes of the meeting. For example, I discovered that the order of the agenda items was crucial: The discussion went much more smoothly if I put the easy items first so we had clearly accomplished at least something by the time the meeting ended.

The more difficult items often could be discussed indefinitely, as everyone realized, and by putting them at the end I could impose a deadline, which helped to focus the discussion. Designing the agendas properly usually allowed me to respect a one-hour rule: Sixty-one minutes spent in a meeting was often one minute too many. (Although even here, the Abraham Lincoln length-of-legs rule applied: If a meeting really required more time, more time it got.)

Designing a visual presentation is in some ways like designing the agenda for a meeting. You need to organize it so that you can ease the audience into the flow, and you need to know how to end it crisply. However, unlike many meetings, a presentation should have a single overarching theme and grand mission, supported by all its parts. The theme—the "Big Picture"—should be explicit from the outset, and you should use it to organize all that follows.

In this chapter we consider how to prepare a presentation. From the outset, you need to keep in mind that you are preparing a *presentation*—not an article, not a written report. Apparently, the United States military recently has not always made this distinction and actually used PowerPoint presentations in lieu of formal battle plans! This is an error. A presentation lacks the detail and precision of a written report, which makes it inappropriate for this role. But if you want to engage the audience and clearly convey key ideas and a handful of specific points, then a multimedia PowerPoint presentation can be just what you need.

8 Guidelines for the Overall Structure of Your Presentation

Eight is an important number, and not just because of the eight psychological principles I've summarized. The Buddhists speak of the "eight-fold path to enlightenment," there are eight cardinal directions on the compass, icons on a computer screen traditionally are multiples of eight pixels wide, and so on. In the following section I'm going to suggest another reason why eight is an important number. I'll offer eight guidelines for building presentations that communicate your message effectively.

1. Prepare to speak to a particular audience.

To make your message relevant to your audience, tailor your presentation to their knowledge, beliefs, and goals. To do so, you must find out what information is important to them and which level of presentation is appropriate for them. I was at a wedding reception recently, and

while we were anxiously waiting to attack the huge and sumptuous cake, one of the guests gave a 15-minute toast, detailing how the happy couple met and what each of them said during their first date, and even a few embarrassing details of what they thought *after* the first date. For most of the other guests, this was TMI (Too Much Information).

How can you get to know your audience before you prepare your presentation?

- In many cases, this is easy: The audience members belong to a specific group, which almost by definition has specific knowledge and interests.
- If you can't fathom their interests from knowing the nature of their group, the first thing to do is simply to ask the events' organizer about the expected attendees. Find out what they do for a living, what sort of background they have, and what they are likely to want to know.
- What do you do if your audience includes people who have a range of backgrounds and interests? My advice is to aim for the middle range, but add some information (on a slide, or in a paraphrase) specifically aimed at the lower and higher ends. Avoid going for the lowest common denominator, which will only bore most of the audience.

2. Show and tell.

There's a reason that your elementary school teachers had you "show and tell." Different parts of the brain deal with language and with vision, and we humans store the two sorts of memories separately. When you present information both visually and verbally, you have two chances to get that material into the memory banks of your audience's minds.

So, combine graphics and text whenever possible to

- help the audience remember what you are saying.
- engage the audience's attention by providing visual variety. We notice what's different—and don't notice what's the same. So seeing a variety of visuals will keep your audience's interest and attention. Video clips can be even more effective, because of the added dimensions of movement and sound (see Chapter 3).

3. Plan in advance how you will direct the audience's attention.

Controlling what the audience looks at and listens to will maximize the chances that they will follow you. To do so:

- Each slide should contain only as much as you can read aloud or describe in about one minute. If you present too much text or too many graphics, the audience will be looking at one thing while you are saying another.
- Vary the salience of what you show by making what's important perceptually distinct. Capture the audience's attention by making important elements larger, brighter, or louder, so that you control what the audience members pay attention to.
- How you highlight your material should depend on the type of material you are highlighting: If you want the viewers to understand a complex structure (e.g., an organizational chart), it makes sense to build up a slide one part at a time, only showing the part you want to talk about at that moment. If you simply want them to focus on a specific part, it makes sense to "build a pointer" into your slide by including a red arrow that points to the subject of interest, or put a circle around a portion of a graphic or text that is your focus.

4. Don't lose your basic message by providing either too much or too little information.

Keep your eye on the ball: Remember what message you want to communicate, and what the audience members should take away with them when they leave the room. Provide no more and no less information than is needed to accomplish this goal. The usual temptation is to provide more information than is necessary to make your point, which is not a good idea for at least two reasons:

- By forcing the audience to *search for* what is important, you prevent them from devoting their full resources to *processing* it.
- You may tax the audience to the point that they simply tune you out.

In cutting out irrelevant detail, be careful not to discard crucial aspects of your material. Cut to the bone, but not into it.

5. Prepare your slides to function as your notes; don't rely on your memory.

Giving a presentation is anxiety provoking—I speak from experience, having given many hundreds of presentations and still feeling a bit anxious before giving one. Being in this anxious state makes it hard to keep in mind the points you need to make. So:

- Prepare the presentation so that you can use it as a memory aid. Include a slide that states every point you want to make. Use titles and graphics to prompt you to discuss specific ideas.
- The more familiar you are with the material, the more abbreviated these prompts can be. But don't abbreviate the material on the slides to the point that it won't make sense to the audience. The slides that prompt you should also underline key points for the audience to help them easily comprehend and remember.
- Yes, the PowerPoint program does provide tools that allow you to see notes that are not projected to the audience, but I advise that you avoid using them, for two reasons:

  1. You need to be relatively near your computer's screen to read them. And because we humans are incredibly good at spotting the target of another person's gaze, the audience will see you looking down at your computer, not at them or at the image projected on the screen that they see. The audience will tend to look where you look, so they will be looking at the back of your computer, rather than at the screen. The audience might also feel that you are ignoring them.
  2. By not sharing the information you are using to give the presentation, you may be saving the best stuff for yourself. Let the audience use your prompts to help them organize and remember what you show and tell.

6. Use the full range of communication options.

Some types of material or topics do not lend themselves well to bulleted lists or even graphics and videoclips. For example, there is a famous parody of a PowerPoint presentation in which the Gettysburg Address is reduced to six slides (see www.norvig.com/Gettysburg/sld001.htm). This medium is not meant to replace other forms of communication but rather to add to them. Telling a personal anecdote that has a crisp bottom line may be far more effective in some cases than listing the key points—or even presenting an illustrated version. In these cases, use the PowerPoint tools to

- structure your presentation.
- allow the audience to see how what you present fits into the structure.
- prompt you to present the anecdote by providing a title, such as "An Unlikely Story and a Moral," perhaps with an appropriate graphic, to cue you to tell your tale.

7. Build in breaks that allow the audience to "come up for air".

In the South of France, home of foie gras, the farmers put funnels down the throats of ducks and force-feed them until their livers grow large and fatty. The ducks probably don't like this treatment, and audiences definitely don't like being given the equivalent in a presentation. So, give the audience time to digest what you've said. You can do this in two ways:

- Break up the information with a cartoon, joke, or demonstration. But if you use a joke, be careful: First, as the vampire bats showed you in Chapter 1, the joke must relate to the topic of your presentation. Second, avoid jokes about politics, race, or religion (your mother probably told you this, and she was right). Avoid making fun of your audience. Lawyer jokes are almost always safe—unless, of course, you are speaking to lawyers.
- Show an entertaining video clip that illustrates your point by reinforcing information you've already provided.

8. Prepare for questions.

One of the worst fates a speaker can suffer is when the question period at the end is met with dead silence; we all know that this does not show that the audience was bowled over and is now utterly convinced. I gave a presentation this year that was followed by an open-ended question period; after an hour and 10 minutes, the moderator cut it off. I considered this vigorous discussion a sign that the presentation was a resounding success: Questions mean that you engaged the audience enough that they thought about the material and want feedback regarding connections they've made or want more information. If you've given your presentation well, you should expect questions.

- Master the material. Be able to explain exactly what is on every slide.
- Anticipate questions that might come up in the question period and reserve some slides to help answer them. (You no doubt will have more to say than you can fit into your presentation and thus can use the question period to slip in additional points.) The hyperlink feature of the PowerPoint program lets you turn any object on a slide into a button link to any other slide. So, you can define a button on the final slide in advance, to take you to your reserved slides whenever the audience is ready for them.

- If nobody immediately steps up to the plate with the first question, be patient. If you are still standing expectantly with a pleasant expression on your face after 20 or 30 seconds and there still are no takers, you might make a little joke such as "I hope I wasn't skating fast on thin ice" or "Should I be taking this as a testament to my powers of persuasion?" But don't fear: Social pressure alone will produce the first question, and more will follow.

The previous eight recommendations apply to the presentation as a whole as well as to each of its sections—the introduction, body, and wrap-up. Now, we'll turn to guidelines that apply to the individual sections themselves.

Building the Introduction

An effective presentation has a clear structure, with each part's having a distinct focus that clearly relates to the other parts. Think of the overall structure of your presentation as a pair of bookends flanking a set of books: The first bookend is the introduction; the books themselves are the body of the presentation; and the final bookend is the wrap-up. We begin at the beginning, with the introduction. An important fact about human memory is that we remember best the beginning and end of a sequence, so the beginning and end of your presentation are particularly important.

To illustrate the power of the principles, I'll give examples of key parts of a presentation I've prepared on how to construct effective graphs.

1. Start with a bang.

A president of the United States may have a 100-day honeymoon period after taking office, but as a speaker you'll be lucky if you have even a five-minute grace period after taking the podium, during which the audience is willing to give you the benefit of the doubt and listen. If during this period you don't convince the audience that you have something of value to say, you will be likely to lose them.

- Your very first slide should define the topic and set the stage for your presentation. To do so, it can simply provide a title, your name, your affiliation, and any other vital information. (For details about titles, see Chapter 3.)
- One way to construct an effective opening has two distinct phases:

1. Prepare a graphic slide that not only defines the topic, but also sets the emotional stage; this slide can be visible before you take the podium, greeting attendees when they walk into the room. For example, if you are talking about dwindling oil reserves, you could use a picture of oil fields burning in Iraq.
2. Prepare a second slide that is identical to the first, except that the background is very low contrast and the title of your presentation, your name, and your affiliation are superimposed in clearly discriminable type. To begin your presentation, you can fade the first slide into the second (using the fade transition that is available in the PowerPoint program), which is an effective way to grab the attention of the audience so that they notice the additional information. Be sure the background of the second slide is very low contrast, to make the text stand out.

Example: Figures 2.1 and 2.2 are adapted from a presentation I've given on the principles summarized in Chapter 1, as applied to designing graphs per se. In the first slide, I provided the title of the presentation and my affiliation; given the topic, I opted not to begin with a fancy graphic—a clean and simple look is most compatible with the subject matter. In the second slide, I presented an example that both grabs the attention of the audience and provides a very general orientation to the topic ("Les Papillion D'Anna," http://kss.free.fr/0kssgallery.htm, email: kss@free.fr). I point out that computers now allow people to make displays very easily, for better or worse. I also point out that what I'll be talking about are all psychological issues—many of which have to do with how we perceive and understand what we see. I note that if the point were to communicate, there would be three problems with the slide I present: (1) It is difficult to tell what is being shown because the contrast is poor; (2) it is hard to organize the image into coherent shapes because the colors are not sharply defined; and (3) moreover, without knowing the context, it is difficult to know why certain features are included while others are not. What do those birds mean, anyway? And is that really a giant squid in the background? If nothing else, this initial exercise intrigues and draws in the audience.

2. Build on the audience's knowledge and concerns when you explain why your topic is important.

The next slide or slides should tell the audience why they should pay attention to what you have to say. These slides should define the topic of your presentation, and then set you up to explain why your

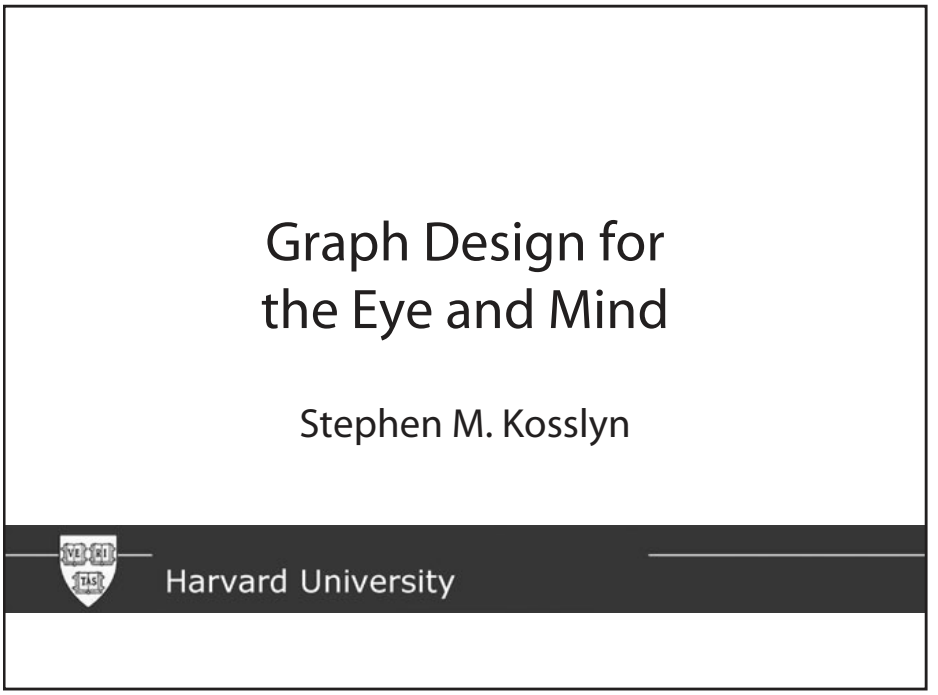

Figure 2.1. A title slide, which conveys the key information about the topic and presenter. Please note that the illustrations in this book can only approximate how slides will look when projected by a particular projector. So there is no substitute for actually viewing a presentation as it will appear to the audience before delivering it, and adjusting contrast and color as necessary.

Figure 2.2. An opening slide that engages the audience. Although this is a work of art, one can ask how it would function to communicate a specific message and point out some of the problems with using it in this way. ("Les Papillons d'Anna," reprinted with permission from Kss. See color version in insert.)

topic is important to your audience. Tap into what your audience already knows and believes.

Example: In my presentation, I needed to tell my audience the point (Figure 2.3), and then immediately acknowledge that people sometimes have the *wrong idea* of what a psychological principle is; they think it has something to do with psychotherapy or Freud. To disabuse them of this idea, I prepared a slide with a cartoon of a person on a couch during psychoanalysis, with a large red circle that has a slash through it (Figure 2.4). The slide following the cartoon prompts me to explain what I mean by the term "psychological principle" (Figure 2.5).

3. Define the topic by providing a concrete example.

Following the initial slides, present graphics or anecdotes that illustrate your topic. If need be, use two or three slides, which can contain any type of graphics (including cartoons), and you can also use sound as appropriate (for more detail, see Chapter 4).

Example: In my presentation, I begin by showing the table in Figure 2.6 and ask whether any gender or income-level group

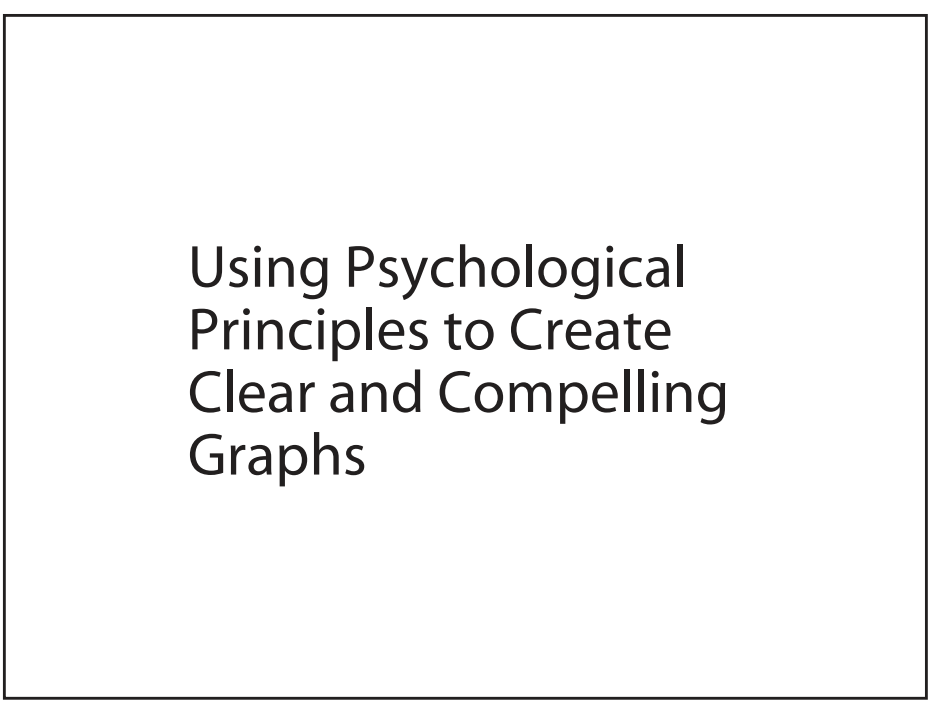

Figure 2.3. This slide introduces the topic of the presentation. The black frame helps to draw the viewers' attention to the text.

Figure 2.4. This slide acknowledges that many people have an overly narrow view of what psychology is and believe that it pertains only to psychotherapy. The audience needs to be disabused of this idea early in the presentation. (See also color version, in insert.)

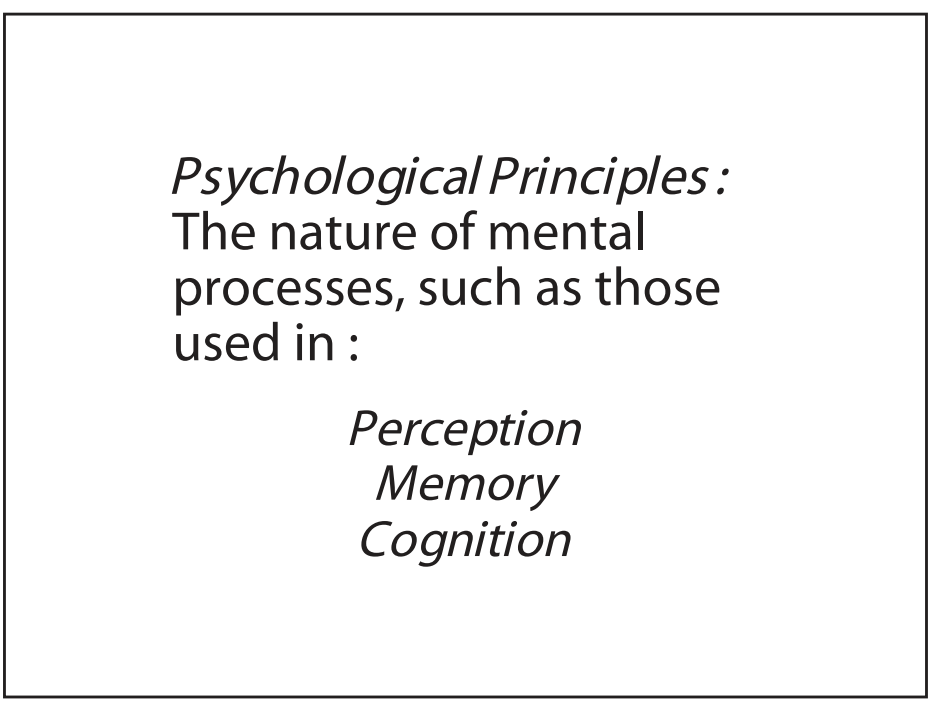

Figure 2.5. After saying what psychology is not (i.e., entirely focused on psychotherapy), I need to add some of the things that it is; this slide prompts me to explain what I mean by the term "psychological principle."

Triglyceride Level by Income Group, Sex, and Age Group

	MALES		FEMALES	
INCOME GROUP	Under 65	65 or Over	Under 65	65 or Over
$0–24,999	250	200	375	550
$25,000+	430	300	700	500

Figure 2.6. To begin to illustrate the power of graphics, I present data in a table and ask whether any group (with group defined by the different combinations of gender and income level) shows different effects of age on triglyceride levels (these are fictional data).

shows different effects of age on triglyceride levels from the other groups. I let the audience ponder the table for four or five seconds (and it's clear from the looks on their faces that they can't tell), and then I present the graph in Figure 2.7. It is immediately evident from this graph that age affects females in the lowest income group in the opposite way from how it affects those in the other groups.

Question: Why is this conclusion so clear in the graph?

Answer: Because our visual systems are very good at detecting differences in the orientations of lines (especially when these differences are greater than 30 degrees of tilt—the difference between a clock's little hand pointing at 12 and big hand pointing at 1), and such differences convey the appropriate information here (which plays to the Principle of Perceptual Organization—specifically, the aspect of Input Channels, as summarized in the appendix).

4. Tell the audience what you want them to conclude.

As part of the introduction, include material that explains what conclusions you are going to support. Prepare slides, video, sound—or

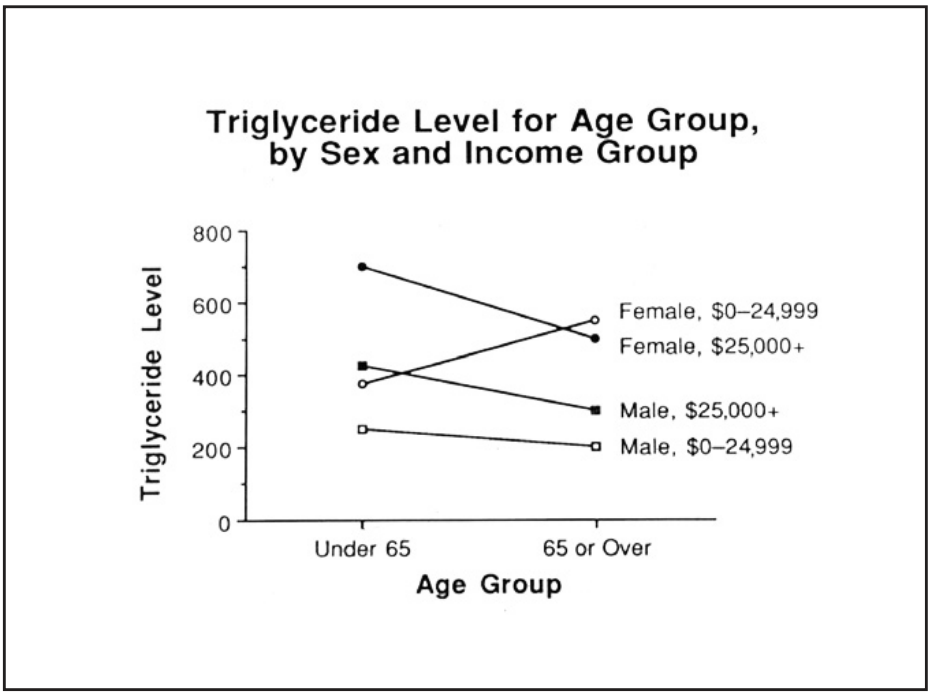

Figure 2.7. This graph presents the same information shown in Figure 2.6, but the atypical group now is obvious at first glance.

just a script for what you will say aloud. That will pique the audience's interest, that will make them wonder how you will get from here to there.

Example: In my presentation, I prepared Figure 2.8 to demonstrate that graphs aren't better than tables in general. I point out that Figure 2.8 displays the identical data as Figure 2.7 but plotted differently; now it's not immediately clear which group deviates from the others. Then—as planned in advance—I ask, "Why not?"

This rhetorical question lets me tell the audience where I am going in the presentation. (The answer, by the way, is that our visual systems are not good at detecting *differences in differences* of height—and that's how the relevant information is conveyed in this particular graph. We humans are terrific at comparing differences in tilt, but not differences in differences of heights. But for all we know a Martian, with a different brain, might have the opposite strength and weakness.)

In setting up this sequence of slides, I've looked ahead to a key part of what I want the audience to conclude: Namely, that our ease in understanding the message in Figure 2.7 versus our difficulty with exactly the same information in Figure 2.8 reflects fundamental facts about how our perceptual systems work, not attributes of the figures or the data themselves.

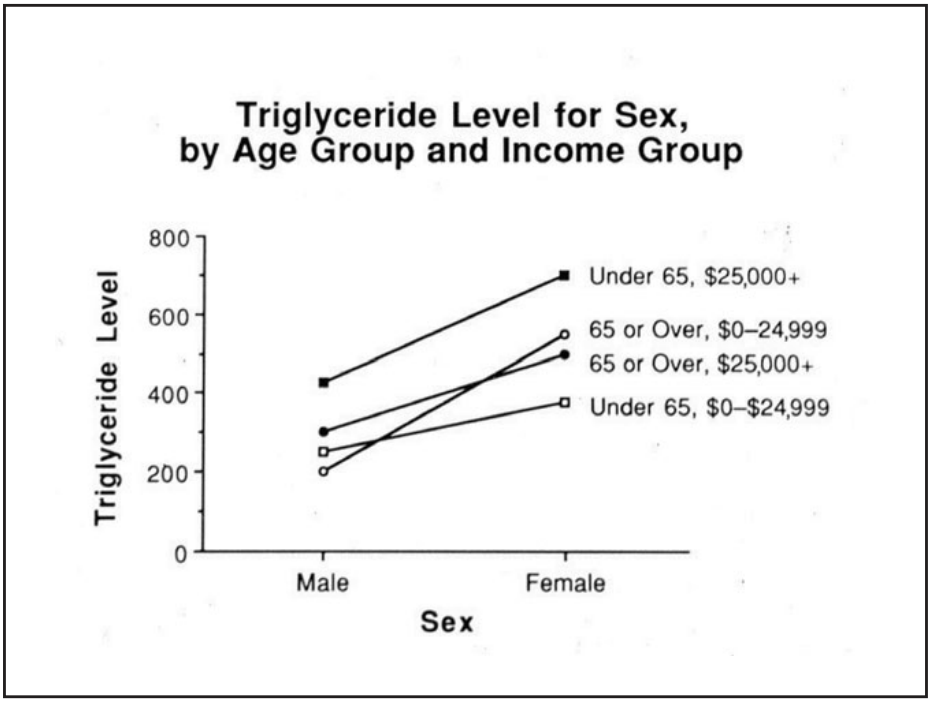

Figure 2.8. This graph presents the same information provided in Figures 2.6 and 2.7, but the atypical group is no longer obvious at first glance. The point: It's not that graphs are always better than tables; graphs are superior only when they play to our perceptual strengths—and avoid relying on our weaknesses. We are good at noticing differences in tilt (Figure 2.7), but not differences among differences in height (this graph).

5. Provide a road map.

If the structure of your argument involves making distinct points or addressing a set of related topics, avoid making your presentation one long string; organize it into digestible modules. And give the audience the conceptual structure of the major parts of your presentation so they can organize what you are going to say. Let them see the agenda. A *brief* outline is a good way to provide a conceptual structure. Organize your outline to reduce the load on your audience members' mental processes as much as possible. As I noted in Chapter 1, a presentation marches on over time; it is not like a written article or report that allows the reader to set his or her own pace and go back when desired. Thus, you need to help your audience follow along so they do not get lost, confused, or overwhelmed.

- The outline should make the structure of your presentation clear.

- Respect the *Rule of Four*. Organize your outline hierarchically into groups of no more than four pieces of information because that's about all we humans can hold in mind at the same time. Each of these groups, however, can in turn include up to four groups (but I don't recommend pushing to the limit—if possible, each group should contain no more than three pieces of information).
- After you present the outline of the major parts of your presentation, briefly explain what is in each part.

Example: In Figure 2.9, I made an outline of the major parts of my presentation on graphs, which focuses on the three goals that should be attained by an effective communication. (These are the same three goals I discussed in Chapter 1, applied to all aspects of presentations.) I explain that these three goals can be achieved by using eight major psychological principles.

Preparing the Body of the Presentation

The following 10 recommendations will help you prepare the body of your presentation.

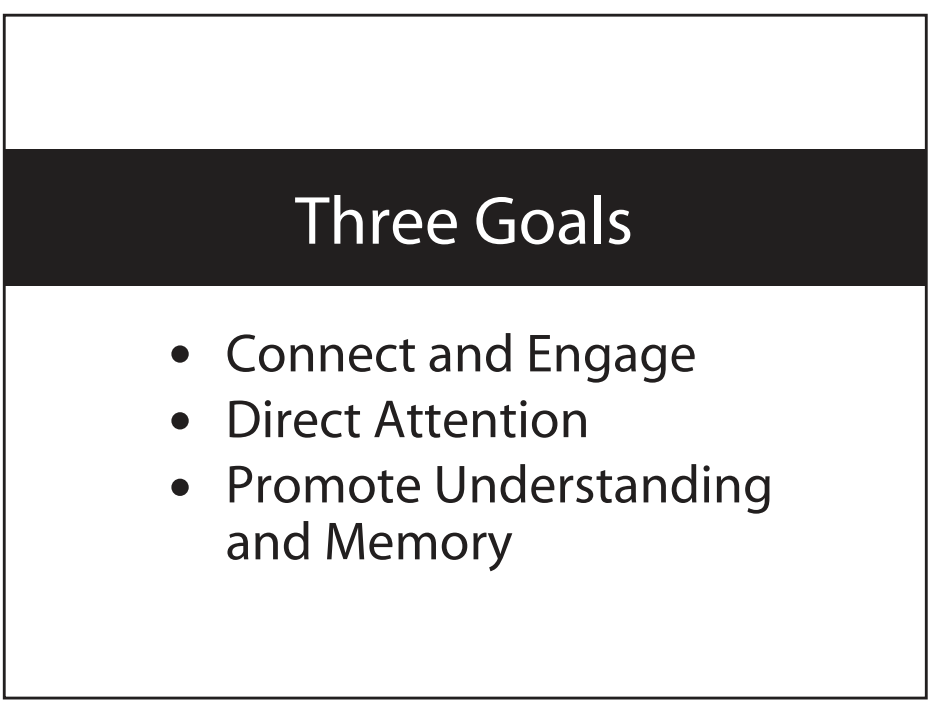

Figure 2.9. An outline of the three goals that will be addressed in the presentation; I briefly explain each one.

1. Tell a story.

The human brain automatically tries to organize and make sense out of experience. One happy consequence of this tendency may be that humans like to tell and understand stories. Take advantage of this proclivity by telling a story in the body of your presentation. In order to do so:

- Create a clear line of argument from the beginning to the end of the body, along which you build a case for the conclusion or conclusions you want to draw.
- Ensure that the transitions between parts are clear.
- Organize the parts so that each provides the foundation for the next.

2. Make a slide that draws attention to the first entry in your outline.

Because you have a story structure, it's important to have the audience members start at the beginning and then work their way through your points, seeing how each one follows from the previous points. Set up your initial slide in the body of the presentation to draw the audience's attention to the first item in your outline. One way to do this is to show the outline with all but the top entry grayed out. Graying out everything but one topic will lead your audience to focus on just the part that is relevant at the moment. Including the grayed out headings will let your audience gauge where you are in the presentation. You can also draw the audience's attention to the relevant part by making this material a different color or a bigger or italic font, have an arrowhead point to it, or make the relevant material slide in from the left.

Example: In Figure 2.10 of my presentation, I have grayed out everything but the topic I want to address in the following portion of the presentation.

3. Prepare an outline for each part that has two or more subparts.

If each of your points needs to be unpacked, then divide each part of your presentation into subparts and introduce them in a mini-outline. Give the audience an organizational structure that will help

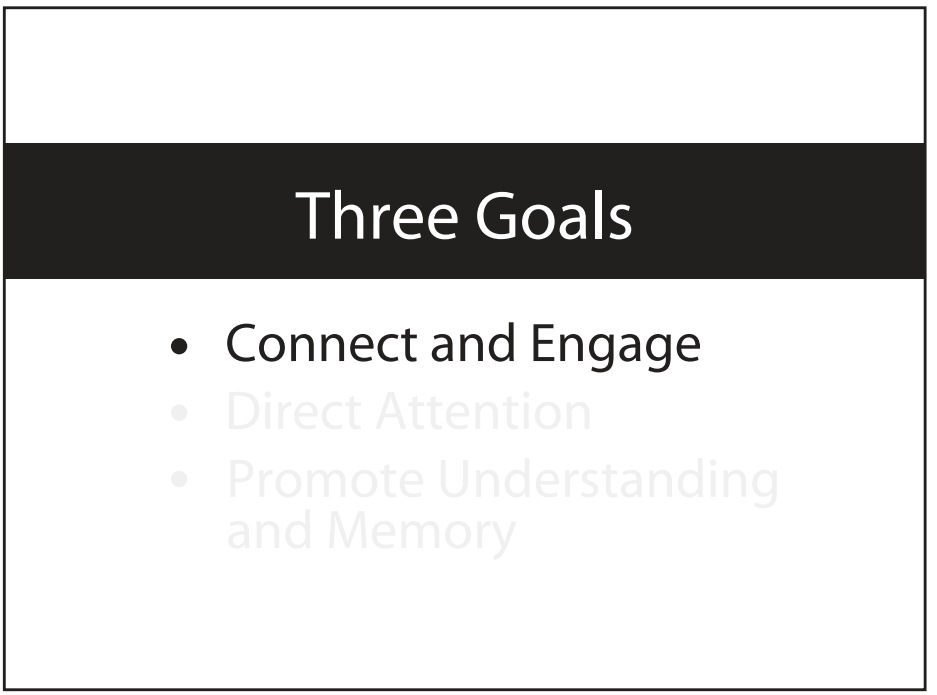

Figure 2.10. The same outline as in Figure 2.9, but with only the top entry salient. By graying out the other entries, I direct the audience members' attention to the relevant heading, which introduces the following section of the presentation.

them understand and remember both your entire presentation and the material you present in each part.

- As indicated in Figure 2.11, show that you're diving more deeply into the outline by reducing the salience of the heading, and by changing the color and the size of the text in the outline or by using different bullets, such as "●" for the first level and ">" for the second level.

  Example: For the main outline, I've used 44-point white type in a black-filled text box for the top-level heading (Figure 2.9), and 36-point black type for the entries. The heading is thus clearly more salient than the entries. For the secondary (subpart) outline (Figure 2.11), I used the 36-point type for the heading, now white in a smaller (and hence less salient) black text box. In general, overall titles should be more salient than entries or titles for subparts. You can also vary salience by varying color. For example, in my presentations, I sometimes use a gray background with a very salient gold for the first-level headings, less-salient white for the next level down, and washed-out yellow for the third level. In all cases, however, the words are clearly discernable from the background.

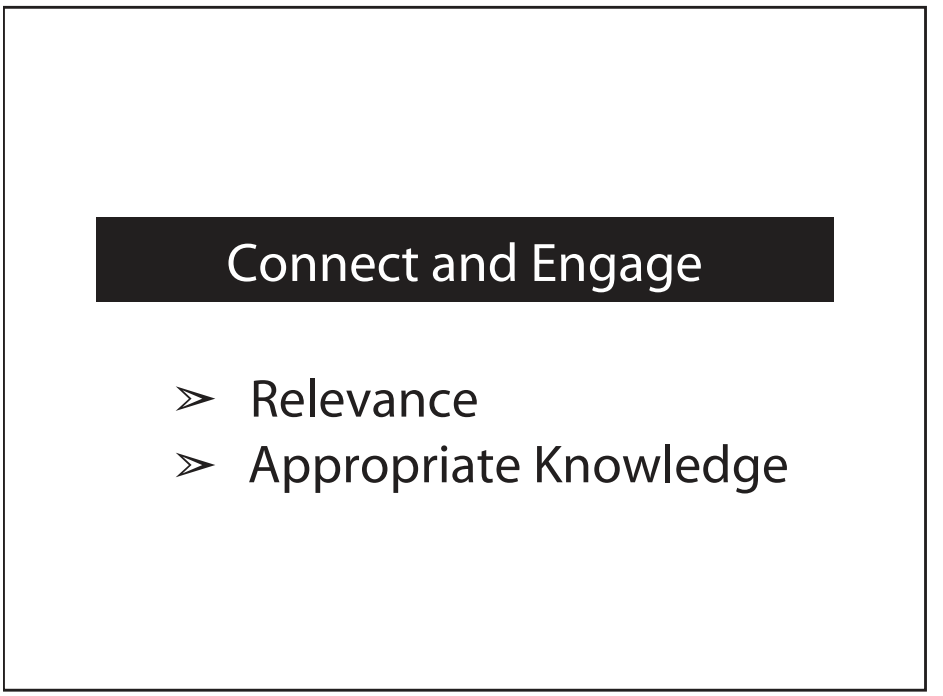

Figure 2.11. Because the following part is composed of two or more subparts, I present an outline of the subparts. As shown here, the names of the subparts should look distinctive and less salient than the name of the part itself.

4. Prepare a slide to draw attention to the first subpart.

After presenting the outline that shows each subpart, ensure that your next slide will draw your audience's attention to the first subpart in the outline.

> **Example:** In my presentation, I used the same technique to draw attention to parts of the presentation as a whole (as illustrated in Figure 2.10) as to its subparts, as shown in Figure 2.12.

5. Define key terms.

I know it sounds trite and obvious, but you should "define your terms." (Do you remember President Clinton's reflections on the word "is," in connection with his unfortunate interactions with Ms. Lewinsky?) Sometimes a lot hinges on exactly what you mean.

- Especially be sure to define technical terms. (If you aren't likely to find it in the front section of *The New York Times*, it's a good candidate for being a "technical term".)

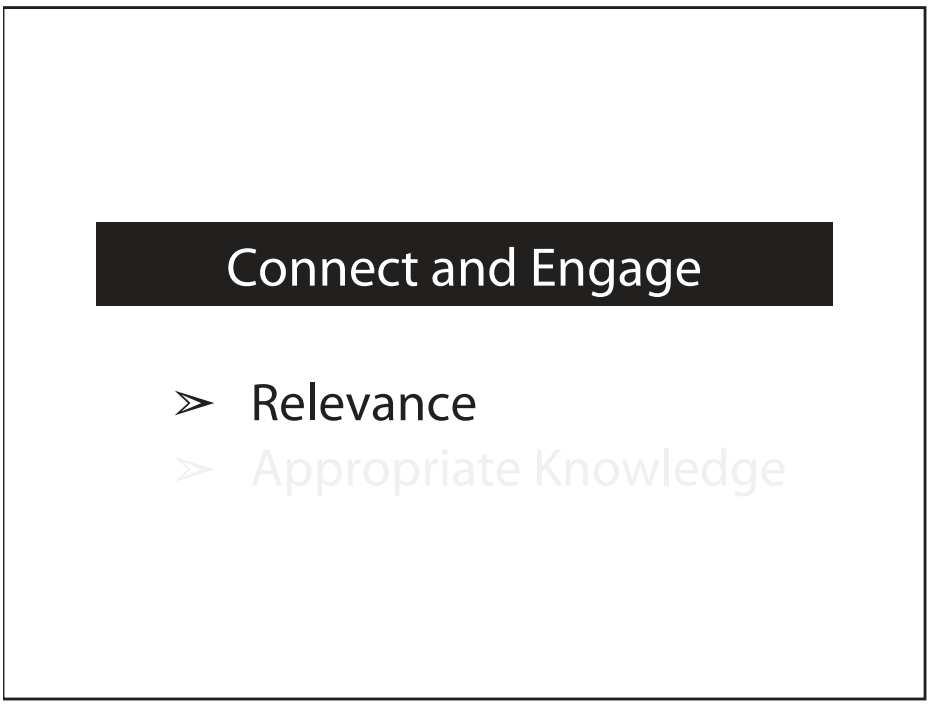

Figure 2.12. I introduce each subpart by graying out the others, leaving salient only the name of the subpart about to be discussed.

- Put the definitions in writing, to let viewers read them as well as hear you say them aloud, which gives the viewers two chances to understand and remember them.

Example: In Figure 2.13 of my presentation, I provide in writing a definition for the first principle I discuss, the Principle of Relevance.

6. Provide concrete, specific evidence to support your conclusions.

As I've stressed, you must decide what message you want the audience members to take away from the presentation as a whole—and you must do the same thing for each part. Think of the story and how it unfolds. Only then can you know how to present clear, specific evidence to back up your message.

Example: In my presentation, I first present an example of when relevance may be violated, in Figure 2.14 (found at www.plotit.com/PLOTIT/PICTURES/GRAPH.GIF), and ask the audience in what circumstances this overwhelming slide would be

Principle of Relevance:

Communication is most effective when neither too much nor too little information is presented.

Figure 2.13. It is important to define key terms in writing (remember: use the slides as your notes), and read the definition aloud. This gives the audience two chances (visual and verbal) to absorb this crucial information.

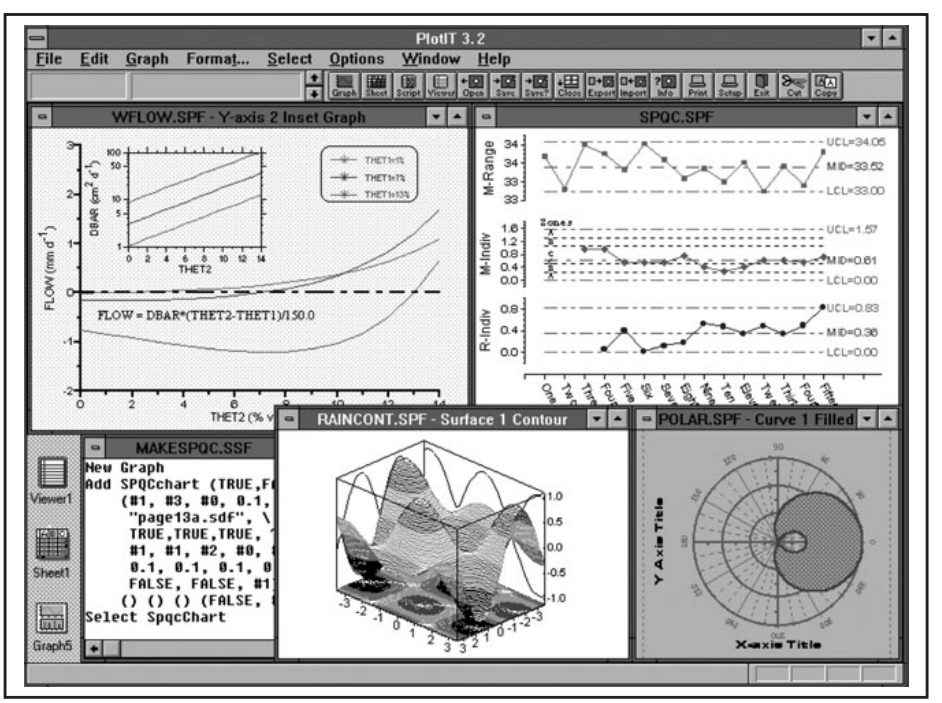

Figure 2.14. If this slide were used to convey specific data, it would provide too much information for the audience to absorb. To drive home this point, you could ask the audience to try to imagine in what situations so much information in such a complex display would be relevant. (Reprinted with permission by Scientific Programming Enterprises.)

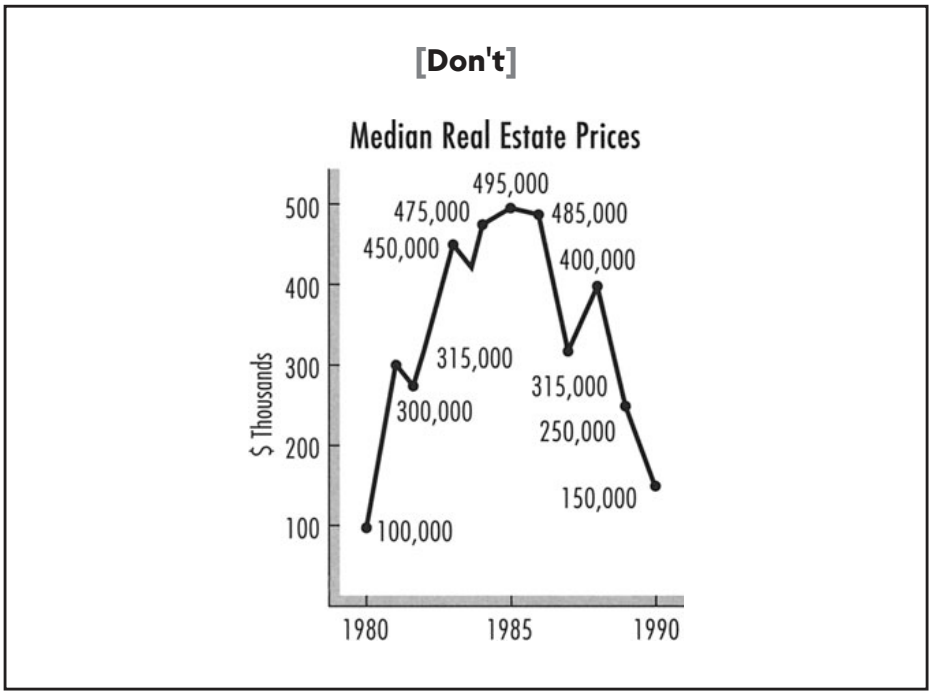

Figure 2.15. This graph illustrates a more common type of display that contains more information than can be absorbed easily. If it is intended to make a particular point, not all of this information is likely to be relevant.

appropriate (e.g., perhaps for marketing purposes, to illustrate what this software can do—we cannot dismiss even this display without knowing the context in which it is used). I then show them an example that presents too much information to illustrate peak and then-current real estate prices, Figure 2.15, and follow with an example that shows only the relevant material, Figure 2.16.

7. Conclude each part and subpart with a summary.

Even if the audience does not follow every word of every part and subpart of your presentation, a summary at the end of each part and subpart will give them the thrust of your presentation.

8. After each part or subpart, highlight the entry for the next part or subpart in the outline to signal its beginning.

Prepare a slide that will draw your audience's attention to the next topic on the outline.

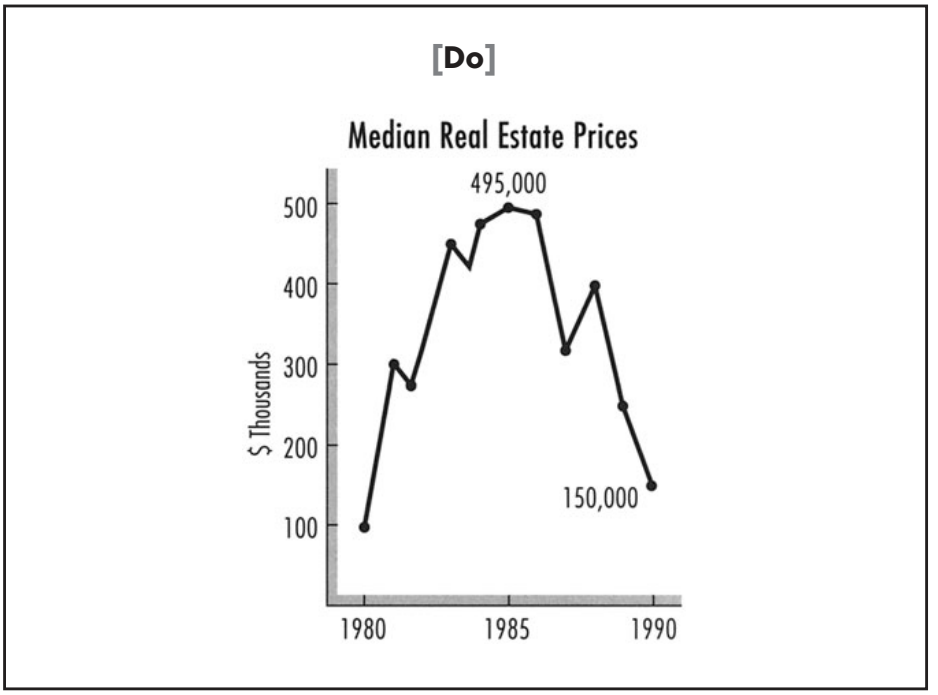

Figure 2.16. This graph presents only the information in Figure 2.15 that is relevant for the discussion, sparing the viewers from having to search for this information.

Example: In my presentation, I returned to the outline of the subparts, shown in Figure 2.11, and then I grayed out the top entry, leaving only the upcoming topic clearly shown. Similarly, after finishing that topic, I summarized it and then returned to the main outline, Figure 2.9. I then grayed out all but the heading for the following topic. Before continuing on to discuss this material, however, I inserted a transition slide, Figure 2.17, to remind my audience that the principles build on one another when they are used in the service of achieving certain goals.

In my actual presentation (not shown here), I then worked through the relevant subparts, which in turn explained and illustrated the Principles of Salience, Discriminability, and Perceptual Organization. Next, I returned to the main outline, and then grayed out all but the third goal. Following this, I presented an outline of the subparts, shown in Figure 2.18. This outline introduced the three relevant principles, the Principle of Compatibility, the Principle of Informative Changes, and the Principle of Capacity Limitations.

Let's pick this up at the end of the third subpart, where I turned to the Principle of Capacity Limitations, as shown in Figure 2.19.

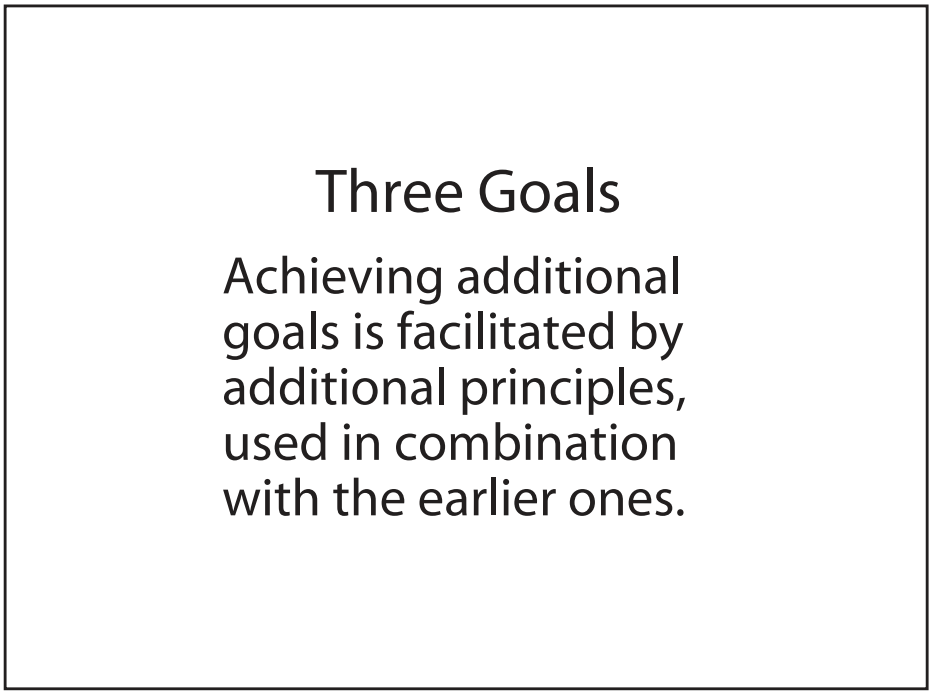

Figure 2.17. Before moving to the material in the next major part, I present a transition slide to remind the audience that principles in the service of the first goal also help to accomplish the following goals.

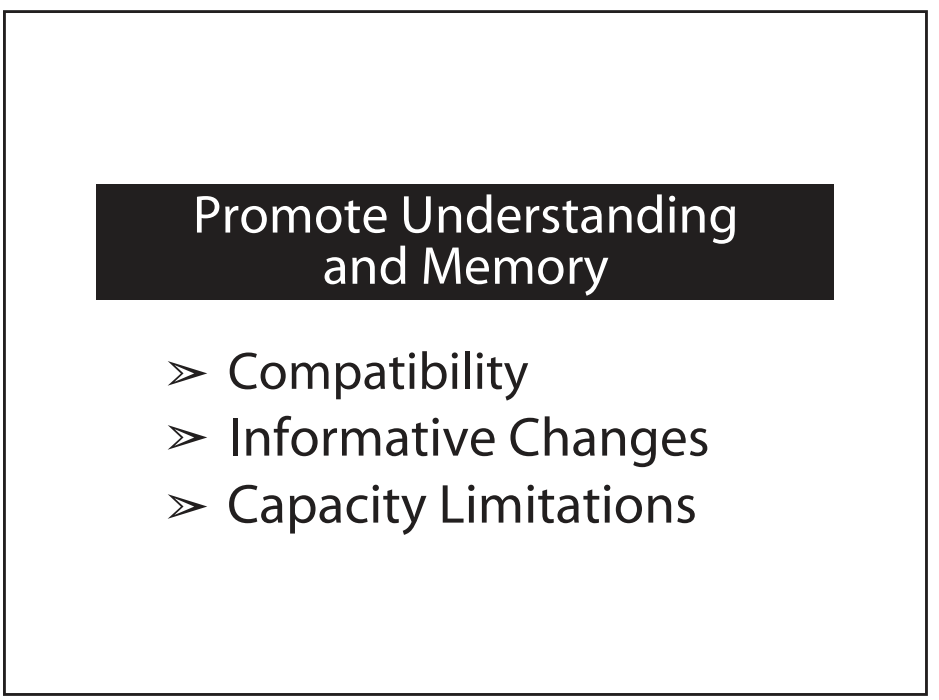

Figure 2.18. This slide shows the outline of the three subparts for the final major part of the presentation.

I defined the principle, Figure 2.20, and turned to one aspect of it, short-term memory limitations, in Figure 2.21.

9. Prepare demonstrations.

As you saw earlier, people automatically pay attention to things that are clearly different. And there is nothing more different than being asked suddenly to shift from being a passive listener to being an active participant. Plus, as you now know, actively processing information helps people understand and remember.

- To keep your audience alert and engaged, prepare a demonstration or even simply ask them to raise their hands if they agree (or disagree) with a specific point.

Example: In Figures 2.22 to 2.26 I have a demonstration of the Principle of Capacity Limitations, focusing on short-term memory limits. Here's how I set it up: When I present Figure 2.22, I read the slide aloud, telling the audience that I'll read a set of directions, such as the ones they are about to see. Then I show Figure 2.23,

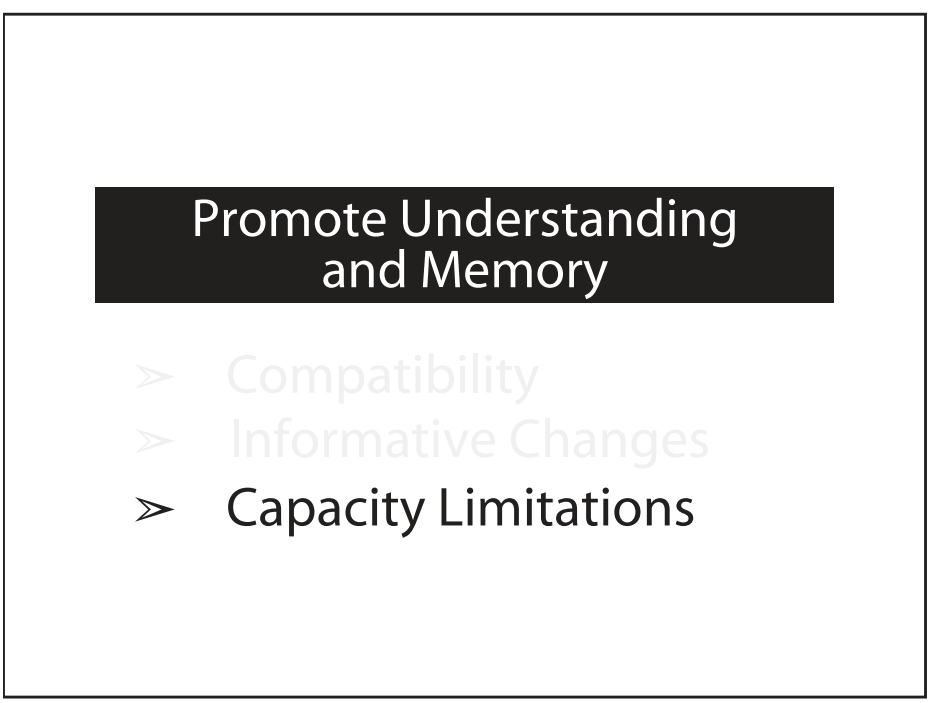

Figure 2.19. After presenting the material for each subpart, I revisited the outline of the subparts at its end, and then grayed out all but the heading that introduced the next subpart, working down the entries until the last one, shown here.

Principle of Capacity Limitations:

People have a limited capacity to retain and to process information, and will not understand a message if too much information must be retained or processed.

Figure 2.20. I again define the principle in writing and read this aloud.

One aspect of this principle:

Limited short-term memory capacity: We can hold in mind only about four units at the same time.

Figure 2.21. I now define a key aspect of the principle, and again read it aloud.

Capacity Limitations in Action :
A Demo!

You will soon hear a sequence of spatial directions (such as North, Northeast, Southeast), and should visualize a one-inch line segment for each one, oriented in the appropriate way, and connect each line to the previous one to form a pathway.

Figure 2.22. This slide summarizes the instructions for the demo, which I read aloud.

North

Northeast

Southeast

South

Figure 2.23. This slide illustrates the first step of a demonstration to illustrate key characteristics of our limited capacity short-term memories. I told the audience that I would read the directions one at a time, and they should visualize a one-inch line segment pointing in each direction, with each successive segment attached to the tail end of the previous one to form a pathway.

Figure 2.24. This slide illustrates the path that the audience should have visualized after receiving the directions in Figure 2.27. I used this and the previous slide to teach the audience the task and to give them a little practice before turning to the actual demonstration in the next two slides.

Northeast

Southeast

North

East

South

East

North

East

South

Northeast

Southeast

Figure 2.25. The actual test directions I used in this demonstration. The audience members were asked to close their eyes and visualize a line segment pointed in each direction, with each line in succession connected to the previous one to form a single pathway.

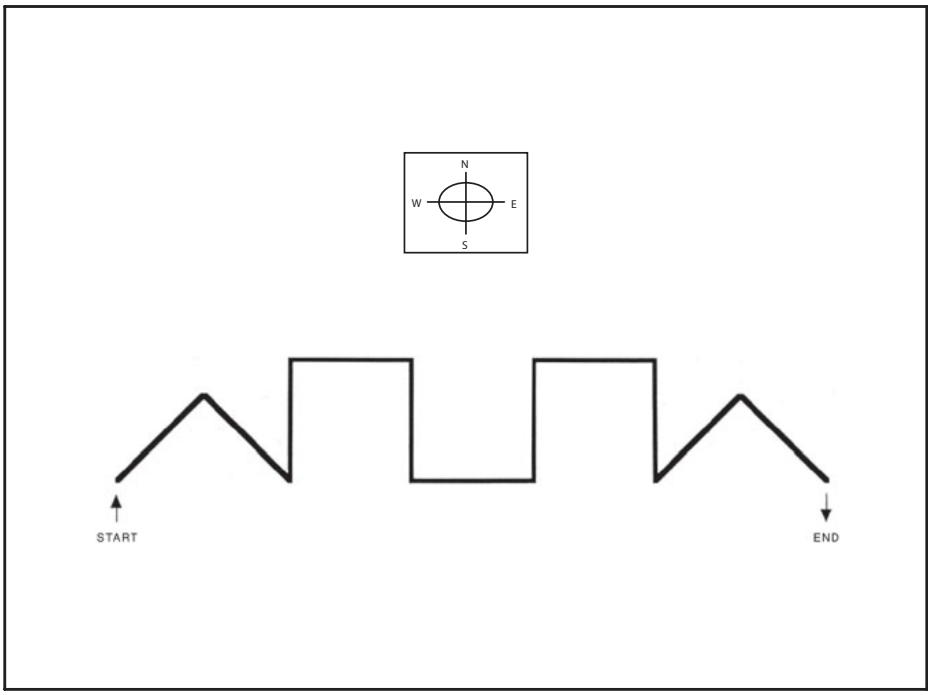

Figure 2.26. The pathway that the audience members should have visualized on hearing the directions in Figure 2.25. Only people who noticed the repeating patterns got this right, which illustrates that we can hold about four units in short-term memory, and that each of those units in turn can be composed of about four elements.

and tell them that I want them to visualize a one-inch line segment for each of these directions oriented in that direction, and to mentally connect each segment to the end of the previously specified one—creating a path of connected segments. Figure 2.24 illustrates the path that would have been formed by following the directions in Figure 2.23. I then ask whether there are any questions about what I'm asking them to do. When the task is clear, I ask them to close their eyes and visualize the pathway while I read aloud each direction in Figure 2.25. I use the PowerPoint program to show me the directions, one every 4 seconds, which is enough time for the audience to visualize the segment.

After reading the last direction, I ask the audience members to open their eyes and look at Figure 2.26 and to raise their hands if they were able to visualize it. Then—and this is the crucial part of the demonstration—I ask them to raise their hands if they noticed the repeating patterns, the peaked and square forms. Finally, I ask how many of those who "got it right" also noticed the patterns. Without fail, virtually all of the people who got it right also noticed the patterns. Why? Because the limit on short-term memory is not

the number of *items* (the individual line segments, in this case)—it's the number of *units*. If audience members could figure out how to organize the segments into four or fewer units, they could store the entire set in short-term memory.

(In the presentation, I follow up this demonstration with a number of examples of graphs that exceed the limits of short-term memory, which I won't present here because I respect the Principle of Relevance—and the overview I've given is good enough for present purposes. In the actual presentation, I show multiple examples, and summarize by returning to the outline in Figure 2.27.)

10. Consider marking your progress with a banner or other signpost.

If your presentation must contain more than four parts or more than four subparts in a part (which, as noted earlier, I do not recommend—three is better still, posing no strain), let the audience know where you are. One way to do so is a banner across the top of each slide. This banner should give the names of the major part and subpart, using the same names as the outline, and make salient the name of

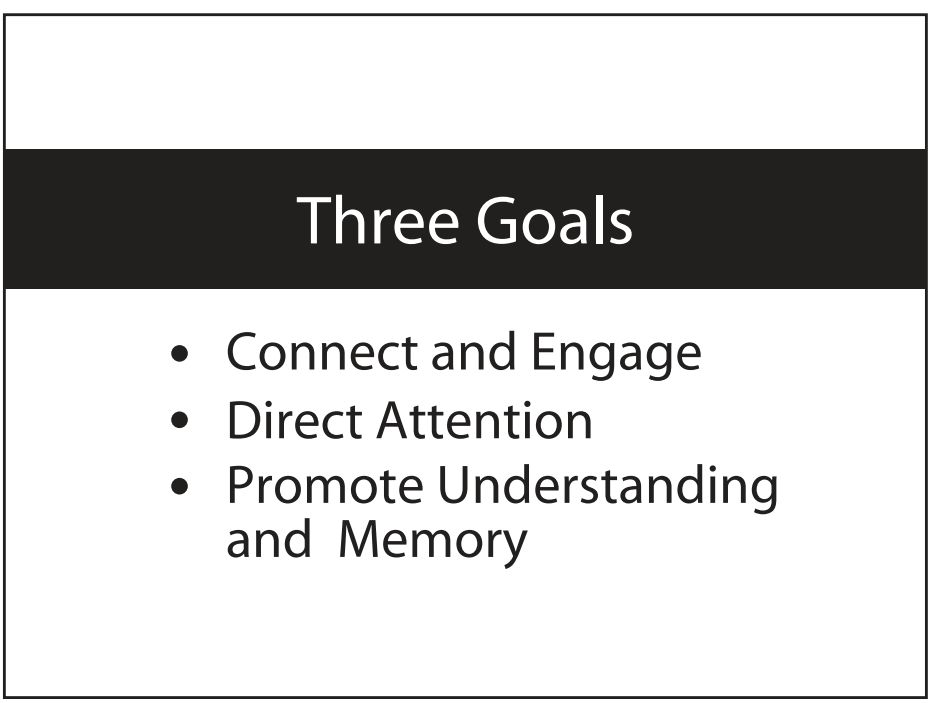

Figure 2.27. At the end of the presentation, I first return to the overall outline and briefly summarize the bottom line reached in each part.

the current topic. An alternative is to include a small box in the upper left or right that keeps track of your progress.

- Ensure that such a signpost is distinct from (i.e., is not grouped with) the other material on the slide by using a different font size or color; you also should make it less salient than the other material on the slide.

I do not include such a banner in my own presentation because many of my slides are so visually complex that they don't leave space (or processing capacity in the audience) for a banner.

Preparing the Wrap-up

My late colleague Roger Brown, whose book *Social Psychology* defined that field, once told me that a good presentation begins by explaining to the audience what you're going to tell them, then telling it to them, and then reminding them what you've just told them. Because people remember best the beginning and end of a sequence, the end of the presentation is very important. Here are three recommendations for the all-important wrap-up of your presentation.

1. Repeat the opening outline.

Make a duplicate of the outline and use it to remind the audience what the presentation included.

Example: As shown in Figure 2.27, show the audience the outline of the major parts and briefly summarize verbally what you said and concluded in each part. In my presentation, I turned back to the eight principles, as shown in Figure 2.28.

2. Prepare text and graphics to emphasize your conclusions.

Ask yourself: If your audience were to give a one- or two- sentence summary of your presentation to a friend, what would it be? Instead of trusting that summary to chance, give it directly. And instead of trusting to memory, prepare slides that make these crucial points.

Psychological Principles:
The Eight-Fold Way

- Relevance
- Appropriate Knowledge
- Salience
- Discriminability
- Perceptual Organization
- Compatibility
- Informative Changes
- Capacity Limitations

Figure 2.28. In this presentation, I introduce the eight principles, and as part of the summary I briefly revisit each of them.

One way to give your audience the message directly is to return to the same graphics that you showed in the introduction, but now add some text or verbal explanation to help your audience see them "in a new light."

Example: In my presentation, I return to Figure 2.2 (repeated in Figure 2.29) and use the principles to explain why it is difficult (assuming that the designer intended it to communicate in the first place!). In an effort to turn a sow's ear into a silk purse, I use the fact that this is not a communication graphic, but rather a work of art, to my advantage—I point out the difference and underscore the importance of relevance.

3. Set up a snappy ending.

You should close crisply, to signal that you've put in place the final bookend. I can't count the number of times I've heard presentations that ended limply with a statement such as, "Well, that's it" or "OK, that's all I want to say," leaving the audience wondering whether more should have been said. An effective way to end with a snap is by using a graphic, or perhaps, even a well-chosen cartoon.

Figure 2.29. To help the audience "see in a new light," I return to the graphic that I used at the outset. I now explain *why* it would not serve to communicate well (assuming that it was ever intended to have that role, which is doubtful—in part because of the problems it would have if pressed into service in this way).

Example: The iconic pizza-pie graph in Figure 2.30 provided the snappy ending for my presentation when I explained it in terms of the principles. (I pointed out that removing one slice lets us distinguish the wedges because it disrupts the good continuation of the boundary, thereby creating another perceptual unit. Then I said, "I was surprised when I first learned that pizza provides a balanced, nutritious meal. I hope this presentation provided the balanced, essential ingredients for your presentations, and thank you for your time and attention."

Another, perhaps better, example: At the end of a presentation on the nature of mental imagery in which I stressed that we can have mental images from more than one of our senses ("seeing with the mind's eye," "hearing with the mind's ear," and so on), I showed a Dilbert cartoon. Dilbert was on a date. In the first panel, his date says, "I cannot begin to describe how I feel when I'm with you." ' Dilbert replies, "Try." In the next panel, she says, "Imagine a spring day, with bright flowers, birds singing and a gentle breeze." Dilbert is smiling. And then comes the last panel, in which she continues: "And now imagine a tractor on your

Figure 2.30. To end on a light note, and also to underscore the generality of the principles, I show this picture of a pizza pie, which serves to illustrate how some of the principles could turn this into a pie graph.

chest." After the audience laughed, I returned to the cartoon and noted the different modalities of mental imagery, the visual (flowers), auditory (birds), and tactile (breeze and tractor). I finished with a comment that the author of the cartoon was not only a keen observer of human relationships but also a perceptive observer of mental processing, having noticed these important aspects of mental imagery. I then thanked the audience for their attention.

Ending on a crisp note like that left a moment of silence, followed by a gratifying amount of applause (probably as much for my treatment of the cartoon as for the talk itself). (Note: If you have any doubt about whether it is appropriate to show or even describe a cartoon, give your lawyer a call—better safe than sued!)

At the very end, repeat your introductory slide (Figure 2.31, in my case), with the title of the presentation and your name—neatly putting in place the second bookend. If you leave this slide on the screen during the question period, it has the added benefit of helping the audience to stay on topic.

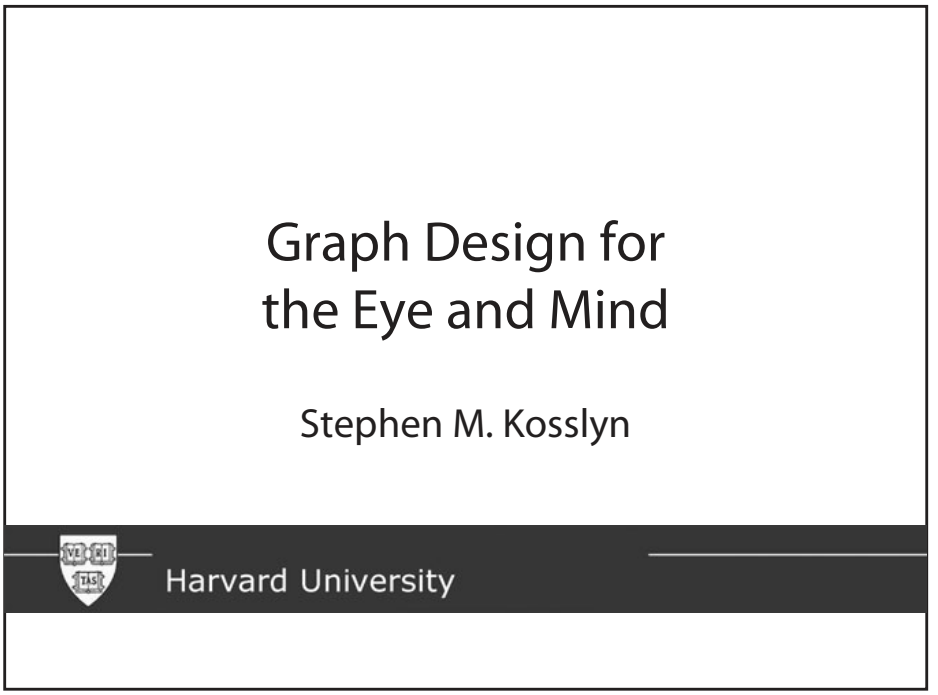

Figure 2.31. I end by returning to the opening slide, which not only reminds the audience of the topic and my identity, but also may help them stay on topic during the question session.

Getting the message across: Delivering the presentation

Because your presentations should be conceived and constructed from the outset to be delivered to an audience, this chapter would not be complete without a brief discussion of how to deliver presentations.

1. Speak to—not at—your audience.

Keep in mind the three following recommendations.

1. Show only key words and phrases; don't show your audience every word you plan to say. I've seen presenters who did this, and they failed to control the attention of their audience—because the audience members were so busy reading that they didn't listen. And in order to fit every word on the screen, the presenter forced the audience to squint to see all that small print.

2. Speak conversationally; don't read your presentation word for word. Yes, what you have to say may require precise wording, and, yes, writing it out in advance and reading it is a security blanket. But these benefits are outweighed by this fact: Unless you've had acting lessons, you'll probably slip into reading in a droning monotone and may soon hear the gentle sounds of snoring. People expect perceptible differences when information is being conveyed; if the sounds stay the same (droning on), the audience members will have trouble staying attentive, let alone awake.
3. Monitor the audience; don't focus your attention solely on your notes and slides. Another problem with reading or focusing too closely on your notes is that you can't easily monitor the audience. You should look at the audience members so that you can vary the speed and level of your presentation based on feedback you pick up. If you catch puzzled looks, slow down and clarify your points; if the audience seems restless and bored, speed up. As Yogi Berra put it, "You can observe a lot by just watching."

2. Face the audience.

Your mother probably told you never to turn your back on people. She was right again; this is good advice and not simply from the point of view of politeness. People can't hear you as well when you talk to the wall, and they can't read your expressions or your lips when they can't see you. And you can't read the audience. So:

- Stand to the side of the screen when you read from it, turning to look out at the audience after reading each point.
- Make sure that you aren't standing in the dark. Dim the lights to the point that your slides are easy to see, but leave enough light for the audience to see your face and gestures (which are another way we communicate, indicating what's most important and most relevant).
- If you expand verbally on material you have presented, move around the stage as you talk, which will keep the audience's attention on you. But stop moving as soon as you present a new slide, video, or audio file. Don't make the audience choose between paying attention to you versus your new material! And never pace, which is always distracting.

3. Don't rush it.

Unless you are the leader of an authoritarian regime (in the Fidel Castro tradition), you probably have a limited amount of time in

which to give your presentation. Thus, you might be tempted to try to cram in as much as possible by talking as quickly as possible. Clamp down on this impulse! Talking quickly and reviewing material quickly can convey the sense of "skating fast over thin ice." Your presentation will be compelling only if it's clear that you have mastery of the material, which is not obvious if you rush.

- Set a speaking pace that lets audience members understand one sentence before you begin to say the next. (Otherwise, you'll overload their limited processing capacity.)
- Give the audience enough time to take in each slide.
- Don't "talk over" the text you show, saying one thing while the audience tries to read another. As I mentioned in Chapter 1, a presentation is no place for a Stroop Effect.
- You don't need to read every slide title aloud if you leave enough time for the audience to read. But do read the most important material aloud to give the audience two ways—visual and verbal—to store it.

4. Know what you can skip.

There's no point in being right on time if you pitched your presentation at too high a level or compressed it too much for the audience, so that much of what you had to say whizzed by them. To avoid either situation:

- Say at the outset that you'll be happy to take *questions of clarification* during the presentation, but would prefer that the audience hold other questions until the end. Such a statement can allow you to stay on time and avoid being derailed by contentious or complex questions. This invitation will also allow you to identify and clarify misconceptions at the outset (it is sometimes surprising how ideas that people bring to a particular topic can lead them to misunderstand what is being said, so it is useful for you to clear the air as early as possible). Moreover, asking the audience to defer general questions immediately after you've invited other sorts of questions softens any sense that you are trying to muzzle them.
- Alternatively, you may be comfortable interrupting the flow and answering all types of questions as they arise, which will make the audience feel that they are being listened to. However, this generous impulse must be managed or the structure of your carefully constructed presentation could be ripped to shreds, which is neither in your nor your audience's interest.
- Keep in mind that you can't always anticipate how long your presentation will take. For one thing, you'll be monitoring the faces and nonverbal signs of the audience and adjusting your pace appropriately. For another, being asked to clarify something will slow you

down. Decide ahead of time what material you can skip through quickly, in case you find that you're running out of time. If necessary, you can tell your audience that you'll go quickly over a section because you've determined that it's less relevant to their needs. Then present the slides in that part very briefly, dwelling only on the concluding slide while explaining what the previous slides led up to.

5. Respond to questions.

As noted earlier, you should have prepared for questions. When you receive them:

- Don't feel that you must answer every question on the spot. Sometimes (rarely) a member of the audience hits on a point that you (the expert) might not have considered. In such cases, if you don't immediately know how to answer, simply say "That's an excellent question. Rather than throwing out my first take and giving you a glib response, let me have a few minutes to think about this." Everyone likes being acknowledged publicly as smart, and this response will come off much better than a rambling, nondefinitive response—which could undermine your credibility in general. In fact, you may even wish to speak to the individual after your presentation and say that you'll need to do some research and will send him or her an e-mail in a day or two to answer. There's no shame in not knowing everything there is to know.
- Always repeat each question as soon as it is asked, so that everyone in the entire room can hear it. Many members of the audience may not hear questions directed at the speaker, particularly if the questioner is near the front of the room. Repeating the questions not only ensures that everyone knows why you'll address a specific point, but it gives you a moment to gather your wits and think about what you should say!
- Close the question period by thanking the audience for their time and attention and for giving you the chance to talk to them. And then say "Thanks again" or make some other positive comment, with a smile and a nod—allowing you to leave the stage on a positive note.

From the Principles to the Point

The eight psychological principles give us plenty of guidance for organizing a presentation. In this chapter you saw guidelines for the

overall structure of your presentation as well as guidelines for the introduction, body of the presentation, and the wrap-up. But in the face of all of these recommendations, I need to emphasize one thing, illustrated by this joke (www.nursinghumor.com/ humor/ lawyer.attorney.legal.a.lawyer.to.the.end.htm):

A lawyer was on his deathbed in his bedroom, and he called to his wife. She rushed in and said, "What is it, honey?" He asked her to run and get the Bible as soon as possible. Being a religious woman, she thought this was a good idea. She ran and got it, prepared to read him his favorite verse or something of the sort. He snatched it from her and began quickly scanning pages, his eyes darting right and left. The wife was curious, so she asked, "What are you doing, honey?" "I'm looking for loopholes!" he shouted.

I remind you that these recommendations are guidelines, not hard-and-fast rules. As long as you respect the eight psychological principles, you can look for loopholes. Here's a review of the recommendations I made in this chapter, in the context of the principles so that you can see where they came from. (I list each recommendation only under the principle that primarily motivated it, although in many cases more than one principle underlies the recommendation.)

Principle of Relevance: Communication is most effective when neither too much nor too little information is presented. (We understand and remember a message more easily when the right amount of detail is used to make the point.)

- Start with a bang; your very first slide should define the topic and set the stage for your presentation.
- Decide what message you want the viewers to take away from the presentation as a whole—and do the same thing for each part and subpart.
- Provide information only when it is needed.
- Don't lose your basic message by providing either too much or too little information; be sure not to swamp the audience with irrelevant detail.
- Include a slide that explicitly states every point you need to make.
- In general, the amount of text you present should just be enough to remind you to present the key points and give the audience a way to conceptualize what you're saying.
- Prepare for questions; hold some slides in reserve, in anticipation of specific questions.
- Don't ignore the audience; monitor them so that you can adjust your presentation.
- Know what you can skip.

- Respond to questions, but don't feel that you must answer every question on the spot.
- Wait 20 or 30 seconds after opening the question session, and if nobody has spoken up, make a self-deprecating comment or ask a rhetorical question (such as, "Was I skating too fast over thin ice?").

Principle of Appropriate Knowledge: Communication requires prior knowledge of pertinent concepts, jargon, and symbols. (We understand and remember a message more easily if it connects to what we already know.)

- Prepare to speak to a particular audience.
- Build on the audience's knowledge and concerns when you explain why your topic is important; find out what information is important to the audience, and which level of presentation is appropriate for them.
- Speak to—not at—your audience; for example, aim for the middle range, with occasional additional information (on a slide, or in a paraphrase) specifically aimed at the lower or higher end.
- Define the topic by providing a concrete example. Explain clearly and concisely why your topic is important to your audience.
- Define key terms, particularly if a term is being used in a technical sense.

Principle of Salience: Attention is drawn to large perceptible differences. (Big relative differences grab attention.)

- Plan in advance how you will direct the audience's attention.
- Capture the audience's attention by making different elements larger, brighter, or louder, so that you have absolute control over what the audience members pay attention to.
- Each slide should contain only what you can read aloud in about one minute.
- Make a slide that draws attention to the first entry in your outline.
- Prepare a slide to draw attention to the first subpart.
- After each part or subpart, highlight the entry for the next part or subpart in the outline.
- To direct attention to specific portions of a slide, build up a slide one part at a time or "build a pointer" into your slide.
- Prepare demonstrations; people automatically pay attention to things that are different, and they remember information best when they process it thoroughly.
- Don't print out your whole presentation and read it word for word (you will probably read in a monotone, which lulls attention).
- If you expand verbally on material you present, move around the stage to keep the audience's attention—but stop moving as soon as you present a new slide, video, or audio file. And never pace.

Principle of Discriminability: Two properties must differ by a large enough proportion or they will not be distinguished. (We need contrast to distinguish shapes, colors, or positions from each other and from the background.)

- Don't show your audience so much text that you will have to make the font very small.
- Face the audience.
- Stand to the side of the screen when you read from it, looking out at the audience after you read each point; they can see the screen better and hear you better.
- Make sure that you aren't standing in the dark.
- Always repeat each question as soon as it is asked so the entire room can hear it.

Principle of Perceptual Organization: People automatically group elements into units, which they then attend to and remember. (These groups are easier to see and remember than the isolated components would be.)

- Ensure that each signpost is clearly distinct from (i.e., is not grouped with) the other material.

Principle of Compatibility: A message is easiest to understand if its form is compatible with its meaning. (For better or worse, the mind tends to judge a book by its cover.)

- Use the full range of communication options.
- If you want the viewers to see how a portion is integrated into a whole, build it up in a series of slides.
- Don't "talk over" the text you show, saying one thing while the audience tries to read another.
- Don't talk quickly and review material quickly, which can convey the sense of "skating fast over thin ice."

Principle of Informative Changes: People expect changes in properties to carry information. (And we expect every necessary piece of information to be indicated by a change in a perceptible property.)

- Use a consistent and distinctively formatted slide to signal the beginning of each new part and subpart.
- Close with a crisp ending, which clearly signals that you've put in place the final bookend.

Principle of Capacity Limitations: People have a limited capacity to retain and to process information and so will not understand a

message if too much information must be retained or processed. (From a communicative point of view, less can be more!)

- Prepare your slides to function as your notes; the slides that you use to jog your memory will also help the audience organize in their minds what you say, so that they can easily comprehend and remember.
- Show and tell: Use graphics in combination with text whenever possible.
- Tell the audience what you want them to conclude; as part of the introduction, explain where you are going in your presentation and what conclusions you are going to support.
- Provide a road map: Organize your outline to reduce the load on your audience members' attention, memory, and reasoning as much as possible.
- Your conceptual structure and outline should be organized hierarchically into groups of no more than four elements.
- Tell a story that helps the audience to organize the parts.
- Provide concrete, specific evidence to support your conclusions.
- Prepare an outline for each part that has two or more subparts.
- Conclude each part and subpart with a summary.
- Consider marking your progress with a banner or other signpost.
- Build in breaks that allow the audience to "come up for air"; give the audience time to digest what you've said.
- Repeat the opening outline at the end, to remind the audience what the presentation was about.
- Prepare text and graphics to emphasize your final conclusions.
- Set up a snappy ending.
- Don't rush it: Don't talk so fast that the audience members are still straining to understand one sentence as you are saying the next, and don't whip through the slides so fast that the audience can't take them in.
- Read the most important material aloud because it will give you the advantage of providing two ways—visual and verbal—for the audience to store what you have to say.